Lesson notes · for you

Fitting a line

Data science · Level 3 · Lesson 1

Time
30 min

Materials
A notebook and a pencil; A tape measure (or a long string and a ruler); A phone or tablet with Darsi, for the lab

Learning goals

- Fit a line to a scatter plot.
- Read the slope and the start.
- Know where a line stops working.

Activity: A ruler line

1. Warm-up (5 min): Look at a classmate and guess their arm span from their height.
2. Plot (10 min): Measure with a partner and add your dots to the board.
3. Predict (10 min): Lay a ruler line, predict a teacher's arm span, then measure it.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- What does the slope tell you?
- Where would the line stop working?
- How close was your prediction?

Check yourself

- mark = 5 + 3 × hours. Predict the mark for 2 hours.

Try it this week
Measure the height and arm span (fingertip to fingertip) of a few people, plot them and lay a ruler where the line fits best. Use it to predict someone new, then measure and compare.

Worksheet

Fitting a line

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Name _____ Class _____ Date _____

1 **The pupils in the data read 0 to 5 hours a week. Should we trust the line for 40 hours a week?**

◯ Only on Mondays ◯ Yes, lines go on forever ◯ No, it is far outside the data

2 **Use mark = 8 + 2 × hours.**

4 hours predicts _____ marks, and the slope is _____ marks per hour.

Word bank: 12 · 16 · 2 · 8

3 **Compute the least-squares slope and intercept, then print the prediction for 6 hours.**

```
x = [1, 2, 3, 4, 5]
y = [10, 12, 14, 16, 18]
mx, my = 3, 14
b = 0
a = my - b * mx
print(round(a + b * 6, 1))
```

Write your program:

4 **A well-fitting regression line proves that x causes y.**

◯ True ◯ False

5 **Data covers 0–5 hours. Which prediction is extrapolation?**

◯ 4.5 hours ◯ 30 hours ◯ 2 hours

✎ **What would you like to predict about your own life? What data would the line need?**

Check your answers

Fitting a line

1 **No, it is far outside the data**

A line only knows the range of data it learned from.

2 **4 hours predicts 16 marks, and the slope is 2 marks per hour.**

8 + 2 × 4 = 16; the coefficient of hours is the slope.

3

```
x = [1, 2, 3, 4, 5]
y = [10, 12, 14, 16, 18]
mx, my = 3, 14
sxy = sxx = 0
for i in range(5):
    sxy += (x[i] - mx) * (y[i] - my)
    sxx += (x[i] - mx) ** 2
b = sxy / sxx
a = my - b * mx
print(round(a + b * 6, 1))
```

b = 20 ÷ 10 = 2 and a = 14 − 6 = 8, so 6 hours gives 20.

4 **False**

Regression measures association; a confounder could drive both.

5 **30 hours**

30 hours is far outside the observed range.